C-HI 8-Node Mini-Production Update

Date: 2026-05-13

Summary

The scheduler-fit `C-HI` mini-production continuation completed cleanly on `kunanyi`. This was a reduced-grid particles-enabled run, not the final full-grid production suite, but it is the first substantially larger production-template continuation beyond the earlier 1-2 node rehearsals.

The run restarted from the previous walltime-stop HDF5 checkpoint at `10.278362 Myr` and advanced to `33.374180 Myr`.

Kunanyi Job

Item	Value
PBS job	`691422.kunanyi-ohpc`
Job name	`c_hi_8mini`
Exit status	`0`
Request	`select=8:ncpus=28:mpiprocs=28`
MPI ranks	`224`
Walltime used	`14:45:00`
PLUTO guard	`-maxtime 14.5`
Restart mode	`-h5restart 7`
Start checkpoint	`data.0007.dbl.h5`, `10.278362 Myr`
Final output	`data.0039.flt.h5`, `33.374180 Myr`
Final restart checkpoint	`data.0012.dbl.h5`, `33.374180 Myr`
Output volume after run	`217G`

The run used the production template staged as `/u/bwbrook/codex_kunanyi/jet_setup_production_template` and reused the existing particles-enabled reduced-grid `C-HI` problem build.

Run Specification Block

This report now records the concrete run specification requested in the 2026-05-13 configuration discussion with Patrick and Stas. A fuller audit is in `docs/science/c_hi_run_configuration_audit_2026-05-13.md`.

Item	Value
Environment	Stewart/Three Hundred tabulated cluster 002003; tabulated_cosmo_002003.csv
Grid preset	kunanyi_single_node_250 reduced-grid rehearsal, not final production grid
Generated grid	404 x 404 x 452 cells; X1/X2 +/-200 kpc, X3 +/-250 kpc
Central cell width	0.05 kpc in all axes
Outer cell width near 200 kpc	X1/X2 about 11.08 kpc; X3 about 8.39 kpc
Physics	HD, Cartesian, particles enabled, JET_RELATIVISTIC_DENSITY = NO
Jet parameters	JET_PWR = 1e45 erg/s, JET_SPD = 0.1, JET_CHI = 1.0, opening angles 15 deg
Injection geometry	JET_INITIAL_RADIUS = 0.1 kpc, JET_INJECTION_HEIGHT = 0.75 kpc, centred launch
Active duration	50 Myr
Output cadence	flt.h5 every 1 Myr; dbl.h5 every 5 Myr; particles every 0.01 Myr
Runtime stack	HDF5/1.14.0-gompi-2023a, OpenMPI/4.1.5-GCC-12.3.0, GCC/12.3.0

Output Coverage

The job added a long block of regular flt.h5 outputs and a new set of restart-capable dbl.h5 checkpoints.

Output type	New useful outputs	Time coverage	File sizes
flt.h5	16-39	10.999144-33.374180 Myr	3.55 GB each; 40 files total
dbl.h5	8-12	14.999293-33.374180 Myr	5.32 GB each; 13 files total
particles	through particles.3343.dbl	about 33.4 Myr	growing to 12.83 MB; 3344 files total

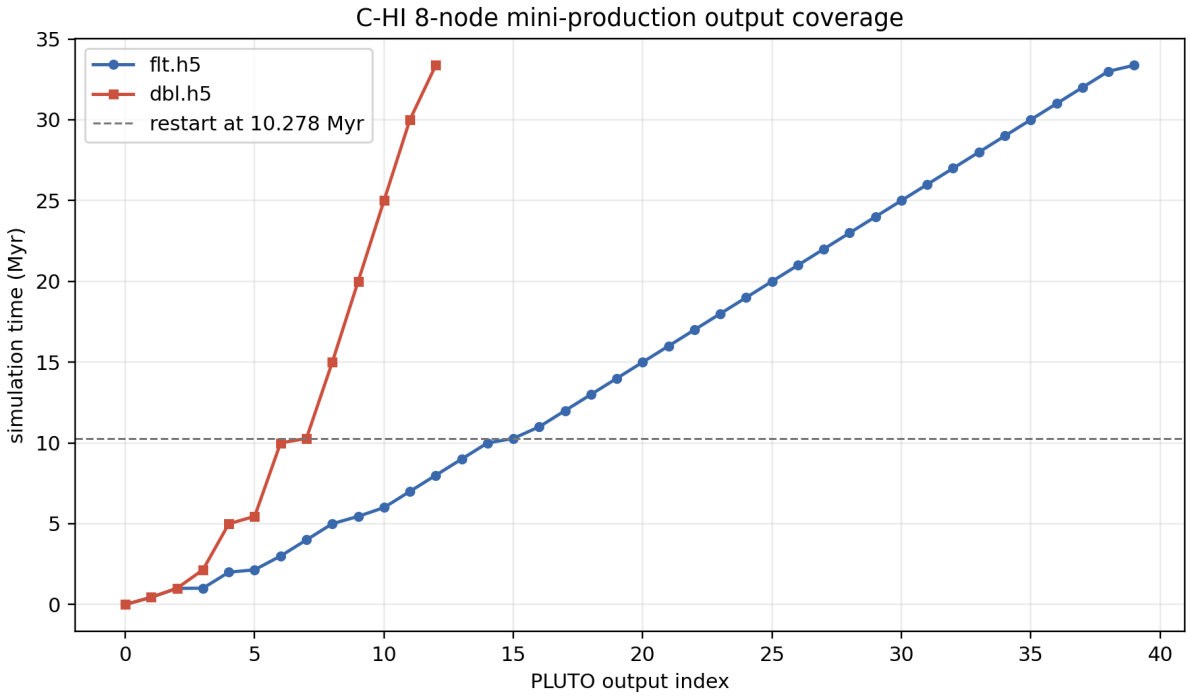


Figure 1: C-HI 8-node output times

The walltime-to-simulation-time throughput for this segment was about `1.57 Myr per wall-clock hour`, measured from `10.278362` to `33.374180 Myr` over the `14.75 h` PBS walltime.

Output Time Table

Selected regular outputs:

Grid output	Time (`Myr`)	Step
15	10.2784	10206
20	14.9993	14817
25	19.9997	19701
30	24.9992	24584
35	29.9996	29468
39	33.3742	32764

Selected restart checkpoints:

Restart output	Time (Myr)	Step
7	10.2784	10206
8	14.9993	14817
9	19.9997	19701
10	24.9992	24584
11	29.9996	29468
12	33.3742	32764

Interpretation For The Meeting

This is a useful operational result:

- the production template restarted cleanly from HDF5
- the 8 node, 224 rank MPI layout ran to the -maxtime guard
- PLUTO wrote the expected final restart-capable HDF5 checkpoint before PBS termination
- output volume is now large enough (217G) that artifact discipline matters
- the run reached two-thirds of the current 50 Myr first-target duration

The result does not replace a later full-grid/full-production allocation test. It does show that the current C-HI setup, restart policy, and HDF5 output path are now boring in exactly the way we wanted.

Follow-Up Diagnostics

Two dependent diagnostic jobs have been submitted:

PBS job	Name	Purpose
691762.kunanyi-ohpc	c_hi_ph5_33	rebuild particle HDF5 through the new particle outputs
691763.kunanyi-ohpc	c_hi_s33	extract truth, cavity, XRAiSE, and radio scalar series

The scalar series targets outputs 15, 20, 25, 30, 35, and 39, covering 10.278-33.374 Myr. Once those jobs complete, the next report should extend the earlier through-10-Myr truth/X-ray/radio comparison to the new mini-production segment.

Local Artifacts

- compact run archive: `build/science/kunanyi_stage_2026-05-08/c_hi_8mini_691422_compact_2026-05-13.tar.gz`
- output-time CSV: `docs/science/assets/c_hi_8mini_2026-05-13/c_hi_8mini_output_times.csv`
- output-time figure: `docs/science/assets/c_hi_8mini_2026-05-13/c_hi_8mini_output_times.png`